

Chapter 9

Geometric Series. Let $r, a \in \mathbb{R}$. If $|r| < 1$, then $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$. If $|r| \geq 1$, then the series diverges.

Ratio Test. Let $\sum a_k$ be an infinite series with positive terms and let $r = \lim_{n \rightarrow \infty} \frac{a_{k+1}}{a_k}$.

1. If $0 \leq r < 1$ then the series converges.
2. If $r > 1$ (including $r = \infty$) then the series diverges.
3. If $r = 1$ then the test is inconclusive.

Root Test. Let $\sum a_k$ be an infinite series with positive terms and let $\rho = \lim_{n \rightarrow \infty} \sqrt[n]{a_k}$.

1. If $0 \leq \rho < 1$ then the series converges.
2. If $\rho > 1$ (including $r = \infty$) then the series diverges.
3. If $\rho = 1$ then the test is inconclusive.

Chapter 10

Taylor's Polynomials. Let f be a function with $f', f'', \dots, f^{(n)}$ defined at a . The **n th-order Taylor Polynomial** for f with its **center** at a denoted p_n , has the property that it matches f in value, slope, and all derivatives up to the n th derivative at a ; that is,

$$p_n(a) = f(a) \text{ and } p'_n(a) = f'(a), \dots, p_n^{(n)}(a) = f^{(n)}(a).$$

The n th-order Taylor polynomial centered at a is

$$p_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

More compactly,

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k \text{ for } k = 0, 1, 2, \dots, n.$$

Chapter 11

Slope of a Tangent Line. Let f be differentiable function at θ_0 . The slope of the line tangent to the curve $r = f(\theta)$ at the point $(f(\theta), \theta_0)$ is

$$\frac{dy}{dx} = \frac{f'(\theta_0) \sin \theta_0 + f(\theta_0) \cos \theta_0}{f'(\theta_0) \cos \theta_0 - f(\theta_0) \sin \theta_0},$$

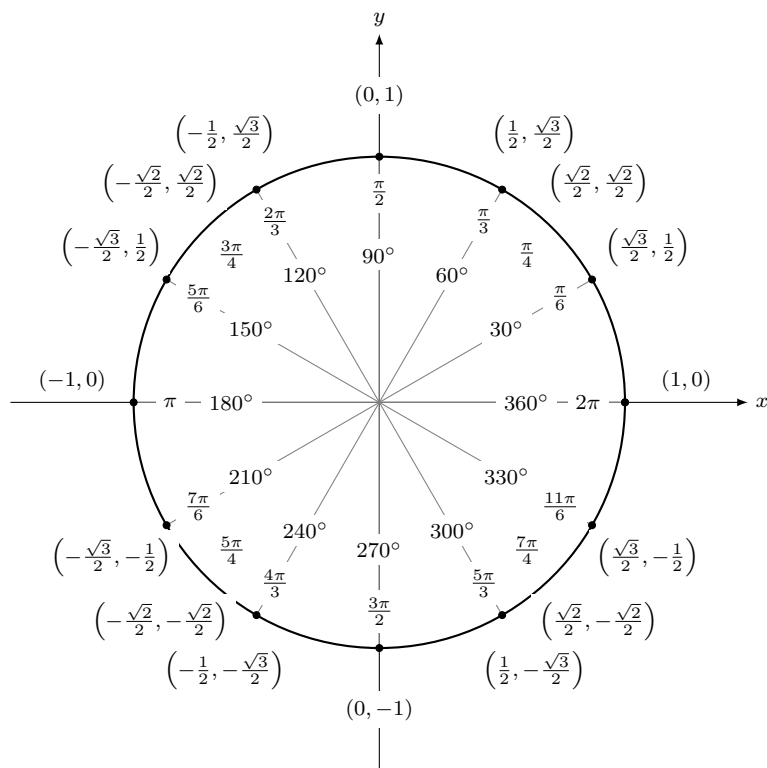
provided the denominator is nonzero at the point. At angles θ_0 for which $f(\theta_0) = 0$ and $f'(\theta_0) \neq 0$, the tangent line is $\theta = \theta_0$ with slope $\tan \theta_0$.

Area of Regions in Polar Coordinates. Let R be the region bounded by the graph of $r = f(\theta)$ and the origin, between $\theta = \alpha$ and $\theta = \beta$, where f is continuous and $f(\theta) \geq g(\theta) \geq 0$ on $[\alpha, \beta]$. The area of R is

$$\int_{\alpha}^{\beta} \frac{1}{2} f(\theta)^2 d\theta.$$

Helpful formulas

1. $\frac{d}{dx}(x^n) = nx^{n-1}$
2. $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$ (Chain rule)
3. $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$
4. $\frac{d}{dx}e^x = e^x$
5. $\frac{d}{dx}\sin ax = a \cos ax$
6. $\frac{d}{dx}\cos ax = -a \sin ax$
7. $\int x^n dx = \frac{1}{n+1}x^{n+1} + C; n \neq -1$
8. $\int \cos ax dx = \frac{1}{a}\sin ax + C$
9. $\int \sin ax dx = -\frac{1}{a}\cos ax + C$
10. $\int \cos^2 x dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$
11. $\int \sin^2 x dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$



Trigonometric Substitution Identities

1. $a^2 - x^2 \rightarrow a^2 \cos^2 \theta$ where $x = a \sin \theta$
2. $a^2 + x^2 \rightarrow a^2 \sec^2 \theta$ where $x = a \tan \theta$
3. $x^2 - a^2 \rightarrow a^2 \tan^2 \theta$ where $x = a \sec \theta$

Pythagorean Identity $\sin^2 \theta + \cos^2 \theta = 1$ **Half-Angle Formulas** $\cos^2 \theta = \frac{1+\cos 2\theta}{2}; \sin^2 \theta = \frac{1-\cos 2\theta}{2}$
Double-Angle Identities $\sin 2\theta = 2 \sin \theta \cos \theta; \cos 2\theta = 2 \cos^2 \theta - 1$